

Module 3

Sinusoidal Steady-State Analysis

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Sinusoids and Phasors

- A sinusoid is a signal that has the form of the sine or cosine function.
- A general expression for the sinusoid

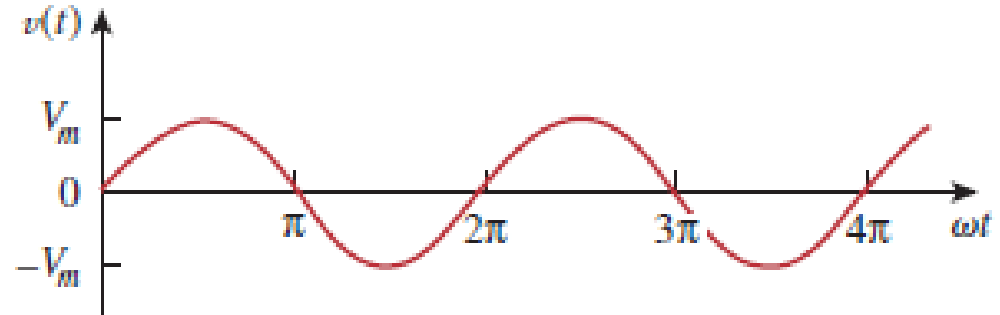
$$v(t) = V_m \sin \omega t$$

Where,

V_m = the amplitude of the sinusoid.

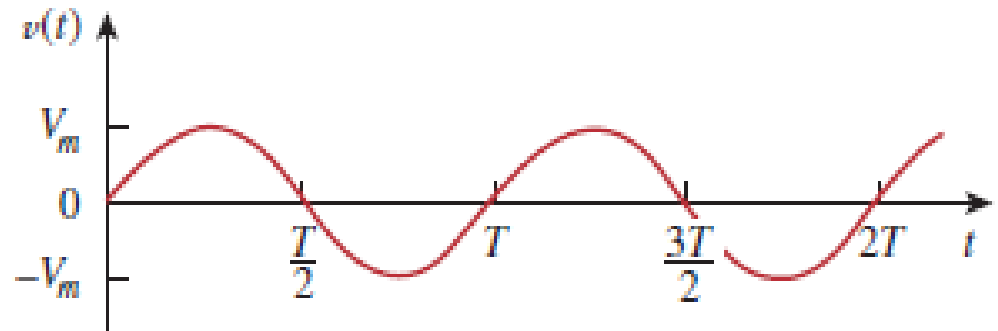
ω = the angular frequency in radians/s

ωt = the argument of the sinusoid



It is evident that the sinusoid repeats itself every T seconds; thus, **T is called the period of the sinusoid.**

we observe that $\omega T = 2\pi$.



$$T = \frac{2\pi}{\omega}$$

Sinusoids and Phasors

The fact that $v(t)$ is repeating for every T seconds can be shown by replacing t by $t+T$.

$$\begin{aligned}v(t + T) &= V_m \sin \omega(t + T) = V_m \sin \omega \left(t + \frac{2\pi}{\omega} \right) \\&= V_m \sin(\omega t + 2\pi) = V_m \sin \omega t = v(t)\end{aligned}$$

$$v(t + T) = v(t)$$

This means $v(t)$ is periodic. In general, $v(t)$ has the same value at $v(t+T)$ *and it is said to be periodic*.

A periodic function is one that satisfies $f(t) = f(t+nT)$ for all t and for all integers n .

The period T of the periodic function is the time of one complete cycle or the number of seconds per cycle.

The reciprocal of this quantity is the number of cycles per second, known as the **cyclic frequency f of the sinusoid**.

$$f = \frac{1}{T}$$

$$\omega = 2\pi f$$

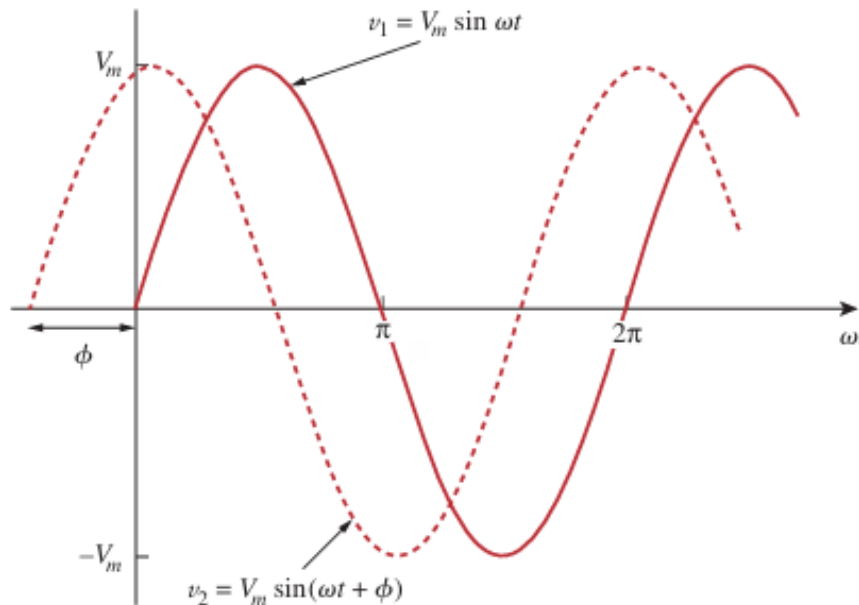
ω is in radians per second (rad/s), f is in hertz (Hz).

Sinusoids and Phasors

Let us now consider a more general expression for the sinusoid.

$$v(t) = V_m \sin(\omega t + \phi)$$

Where ωt is the argument and ϕ is the phase. Both argument and phase can be in radians or degrees. Consider, $v_1(t) = V_m \sin \omega t$ and $v_2(t) = V_m \sin(\omega t + \phi)$



The starting point of v_2 occurs first in time. Therefore, we say that v_2 leads v_1 by ϕ or that v_1 lags v_2 by ϕ .

If we also say that v_1 and v_2 are out of phase. If $\phi = 0$, then v_1 and v_2 are said to be in phase; they reach their minima and maxima at exactly the same time, which means they operate at the same frequency;

They do not need to have the same amplitude.

A sinusoid can be expressed in either sine or cosine form.

Sinusoids and Phasors

Trigonometric identities: Using these relationships, we can transform a sinusoid from sine form to cosine form or vice versa.

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

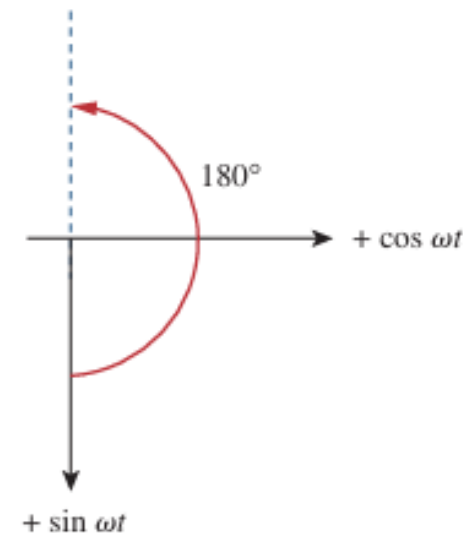
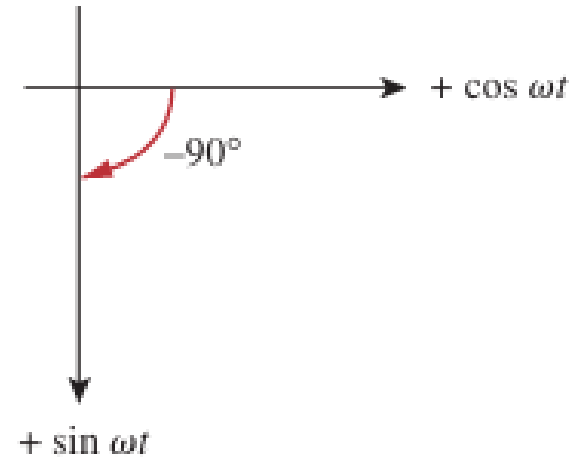
$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

Subtracting 90° from the argument of $\cos \omega t$ gives $\sin \omega t$.



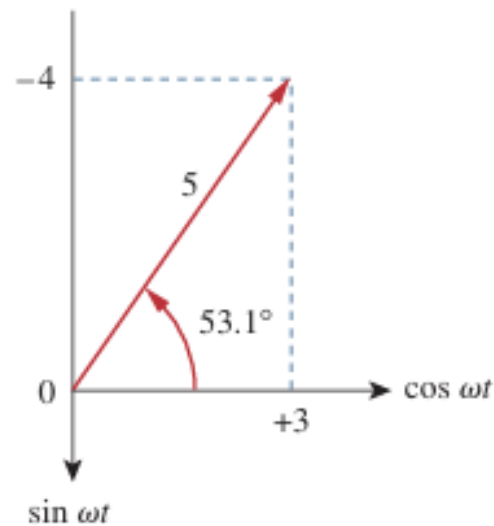
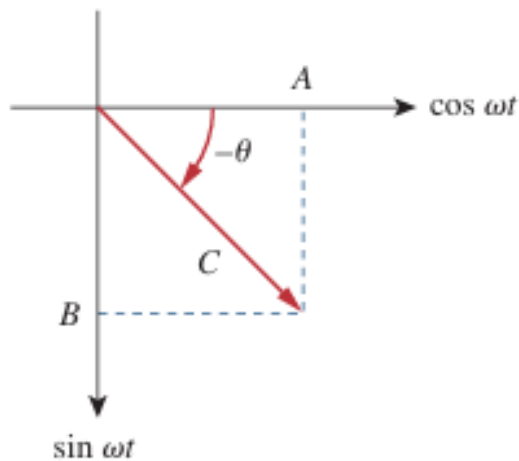
Sinusoids and Phasors

Adding 2 sinusoid $A \cos \omega t + B \sin \omega t = C \cos(\omega t - \theta)$

$$C = \sqrt{A^2 + B^2}, \quad \theta = \tan^{-1} \frac{B}{A}$$

For example,

$$3 \cos \omega t - 4 \sin \omega t = 5 \cos(\omega t + 53.1^\circ)$$



Sinusoids and Phasors

1. Find the amplitude, phase, period, and frequency of the sinusoid $v(t) = 12 \cos(50t + 10^\circ)$

The amplitude = 12 V

The phase = 10°

The angular frequency = 50 rad/s

The period = 0.1257 s

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{50}$$

The frequency is $f = 1/T = 7.958$ Hz.

2. Given the sinusoid $30\sin(4\pi t - 75^\circ)$, calculate its amplitude, phase, angular frequency, period, and frequency.

Ans: 30, 12.57 rad/s, 0.5 s, 2 Hz

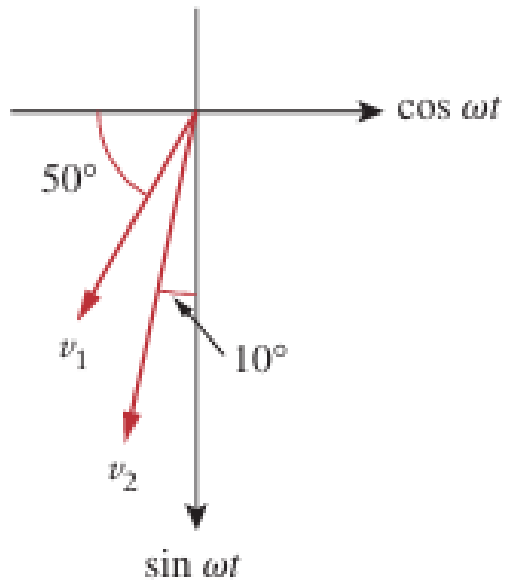
3. Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.

$$v_1 = -10 \cos(\omega t + 50^\circ) = 10 \cos(\omega t + 50^\circ - 180^\circ)$$

$$v_1 = 10 \cos(\omega t - 130^\circ) \text{ or } v_1 = 10 \cos(\omega t + 230^\circ)$$

$$v_2 = 12 \sin(\omega t - 10^\circ) = 12 \cos(\omega t - 10^\circ + 90^\circ)$$

Sinusoids and Phasors



v_2 leads v_1 by 30° .

Sinusoids and Phasors

Sinusoids are easily expressed in terms of **phasors**, which are more convenient to work with than sine and cosine functions.

A phasor is a complex number that represents the amplitude and phase of a sinusoid.

Phasors provide a simple means of analyzing linear circuits excited by sinusoidal sources.

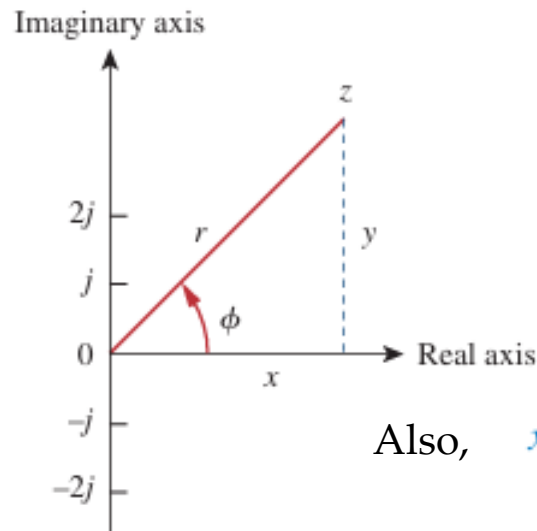
It can be represented in anyone of the 3 forms.

$$z = x + jy \quad \text{Rectangular form}$$

$$z = r \angle \phi \quad \text{Polar form}$$

$$z = re^{j\phi} \quad \text{Exponential form}$$

where r is the magnitude of z , and ϕ is the phase of z .



X axis represents the real part and the y axis represents the imaginary part of a complex number.

$$r = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x}$$

The relationship between the rectangular form and the polar form.

$$x + jy = r \angle \phi.$$

Also, $x = r \cos \phi, \quad y = r \sin \phi$

$$z = x + jy = r \angle \phi = r(\cos \phi + j \sin \phi)$$

Sinusoids and Phasors

Addition:

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Subtraction:

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

Multiplication:

$$z_1 z_2 = r_1 r_2 \angle \phi_1 + \phi_2$$

Division:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2$$

Reciprocal:

$$\frac{1}{z} = \frac{1}{r} \angle -\phi$$

Square Root:

$$\sqrt{z} = \sqrt{r} \angle \phi/2$$

Complex Conjugate:

$$z^* = x - jy = r \angle -\phi = r e^{-j\phi}$$

Note that from Eq. (9.18e),

$$\frac{1}{j} = -j$$

Addition and subtraction of complex numbers are better performed in rectangular form.

Multiplication and division are better done in polar form.

The idea of phasor representation is based on Euler's identity.

$$e^{\pm j\phi} = \cos \phi \pm j \sin \phi$$

$$\cos \phi = \operatorname{Re}(e^{j\phi})$$

$$\sin \phi = \operatorname{Im}(e^{j\phi})$$

Sinusoids and Phasors

$$v(t) = V_m \cos(\omega t + \phi) = \operatorname{Re}(V_m e^{j(\omega t + \phi)})$$

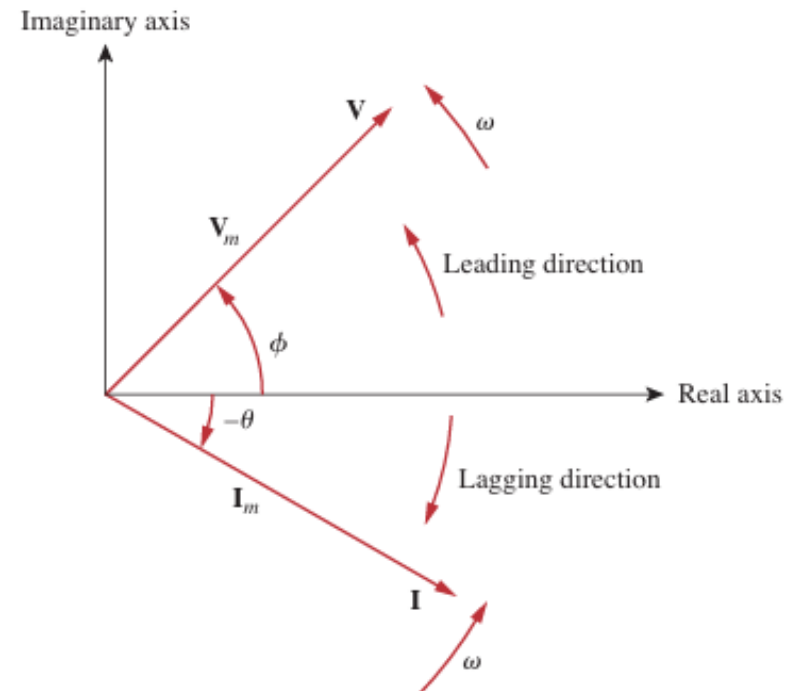
$$v(t) = \operatorname{Re}(V_m e^{j\phi} e^{j\omega t})$$

$$v(t) = \operatorname{Re}(\mathbf{V} e^{j\omega t})$$

Where $\mathbf{V}$ is the phasor representation of the sinusoid $v(t)$.

$$\mathbf{V} = V_m e^{j\phi} = V_m \angle \phi$$

phasor diagram



$$\mathbf{V} = V_m \angle \phi \text{ and } \mathbf{I} = I_m \angle -\theta.$$

Sinusoids and Phasors

Domain Transformations:

$$\begin{array}{ccc} v(t) = V_m \cos(\omega t + \phi) & \Leftrightarrow & \mathbf{V} = V_m \angle \phi \\ \text{(Time-domain} & & \text{(Phasor-domain} \\ \text{representation)} & & \text{representation)} \end{array}$$

Sinusoid-phasor transformation.

Time domain representation

Phasor domain representation

$$V_m \cos(\omega t + \phi)$$

$$V_m \angle \phi$$

$$V_m \sin(\omega t + \phi)$$

$$V_m \angle \phi - 90^\circ$$

$$I_m \cos(\omega t + \theta)$$

$$I_m \angle \theta$$

$$I_m \sin(\omega t + \theta)$$

$$I_m \angle \theta - 90^\circ$$

Sinusoids and Phasors

$$\frac{dv}{dt} \quad \Leftrightarrow \quad j\omega V$$

(Time domain) (Phasor domain)

Differentiating a sinusoid is equivalent to multiplying its corresponding phasor by $j\omega$.

$$\int v \, dt \quad \Leftrightarrow \quad \frac{V}{j\omega}$$

(Time domain) (Phasor domain)

Integrating a sinusoid is equivalent to dividing its corresponding phasor by $j\omega$.

1. $v(t)$ is the instantaneous or time domain representation, while V is the frequency or phasor domain representation.
2. $v(t)$ is time dependent, while V is not.
3. $v(t)$ is always real with no complex term, while V is generally complex.

Phasor analysis applies only when frequency is constant; it applies in manipulating two or more sinusoidal signals only if they are of the same frequency.

Sinusoids and Phasors

Evaluate:

$$(40 \angle 50^\circ + 20 \angle -30^\circ)$$

Transform these sinusoids to phasors:

(a) $i = 6 \cos(50t - 40^\circ) \text{ A}$

(b) $v = -4 \sin(30t + 50^\circ) \text{ V}$

Find the sinusoids represented by these phasors:

(a) $\mathbf{I} = -3 + j4 \text{ A}$

(b) $\mathbf{V} = j8e^{-j20^\circ} \text{ V}$

Given $i_1(t) = 4 \cos(\omega t + 30^\circ) \text{ A}$ and $i_2(t) = 5 \sin(\omega t - 20^\circ) \text{ A}$, find their sum.

Next class: Phasor relationships for circuit elements